StudyPlan AI: Deliverable 1
Problem Definition & Data Foundations

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1 Problem Definition

1.1 Problem Statement

Many students struggle to plan their study schedules effectively based on individual habits and past academic performance. Traditional planners and advice systems provide generic recommendations that ignore personal factors such as study focus, sleep quality, motivation, and lifestyle. As a result, students may underperform or spend study time inefficiently. **StudyPlan AI** addresses this by predicting a student's expected exam performance, identifying areas that need improvement, and generating a personalized weekly study schedule that adapts to each student's behavior patterns.

1.2 Target Audience

The primary users of **StudyPlan AI** are high school and university students who aim to optimize their study efficiency. Secondary users include teachers or academic counselors seeking insights into student performance to provide targeted guidance. This project addresses a research gap by combining academic performance data with behavioral factors to generate actionable and personalized study recommendations.

1.3 AI Justification

Predicting student performance is a complex, non-linear problem influenced by multiple interacting factors such as past scores, study habits, sleep quality, and motivation. Traditional rule-based approaches cannot effectively model these relationships or provide individualized recommendations. Using machine learning, **StudyPlan AI** can predict exam performance and generate counterfactual insights, for example: "If weekly study hours increase from 3 to 6 and sleep hours increase from 5 to 7, the predicted exam score rises from 72 to 85." Furthermore, AI enables automated scheduling that prioritizes areas needing improvement, which would be impractical using static rules alone.

2 Phase 2: Data Acquisition and Preprocessing

2.1 Dataset Collection

The dataset used for this project was obtained from Kaggle.

Dataset: Student Performance Factors Dataset

Source: `https://www.kaggle.com/datasets/lainguyn123/student-performance-factors`

The dataset contains multiple features describing student lifestyle, academic background, and environmental factors that may influence exam performance. Each row represents an individual student record, while columns represent attributes such as study hours, attendance, sleep patterns, parental involvement, and exam scores.

The dataset includes both numerical and categorical variables, making it suitable for demonstrating preprocessing techniques such as encoding, normalization, and feature scaling.

2.2 Dataset Overview

The dataset contains the following key attributes:

- Hours_Studied
- Attendance
- Parental_Involvement
- Access_to_Resources
- Extracurricular_Activities
- Sleep_Hours
- Previous_Scores
- Motivation_Level
- Internet_Access
- Tutoring_Sessions
- Family_Income
- Teacher_Quality
- School_Type
- Peer_Influence
- Physical_Activity
- Learning_Disabilities
- Parental_Education_Level
- Distance_from_Home
- Gender
- Exam_Score

The target variable used for prediction in this project is **Exam_Score**.

2.3 Data Cleaning

2.3.1 Handling Missing Values

Missing values were handled differently depending on the data type:

- Numerical features were filled using the **mean** of the column.
- Categorical features were filled using the **mode** (most frequent value).

This ensured that missing information did not introduce bias or errors during training.

2.3.2 Duplicate Removal

Duplicate records can distort statistical analysis and model training. The dataset was checked for duplicate rows, and any duplicates found were removed to ensure each record represented a unique student.

2.3.3 Outlier Detection and Handling

Outliers were identified using the **Interquartile Range (IQR)** method. Values that fell outside the acceptable range were capped to reduce their impact while preserving the data distribution.

The target variable (Exam_Score) was intentionally excluded from outlier modification to avoid distorting the prediction objective.

2.4 Feature Engineering and Encoding

Machine learning algorithms require numerical input. Therefore, categorical variables were converted into numerical representations.

2.4.1 Ordinal Encoding

Some categorical variables possess a natural order (e.g., Low, Medium, High). These were encoded using ordinal encoding.

Example mapping:

- Low = 1
- Medium = 2
- High = 3

Features encoded using this approach include:

- Parental_Involvement
- Access_to_Resources
- Motivation_Level
- Family_Income
- Teacher_Quality
- Parental_Education_Level

2.4.2 One-Hot Encoding

Nominal categorical variables without inherent ordering were transformed using one-hot encoding. This technique creates binary columns for each category.

Features encoded using one-hot encoding include:

- Gender

- School_Type
- Internet_Access
- Peer_Influence
- Distance_from_Home
- Learning_Disabilities
- Extracurricular_Activities

2.5 Feature Scaling

To ensure that features operate on comparable scales, numerical variables were standardized using **StandardScaler**. This transformation centers the data around a mean of zero with a standard deviation of one.

Scaling prevents features with large numeric ranges from dominating the learning process.

2.6 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns, distributions, and relationships between variables.

2.6.1 Exam Score Distribution

The histogram below illustrates the distribution of exam scores among students.

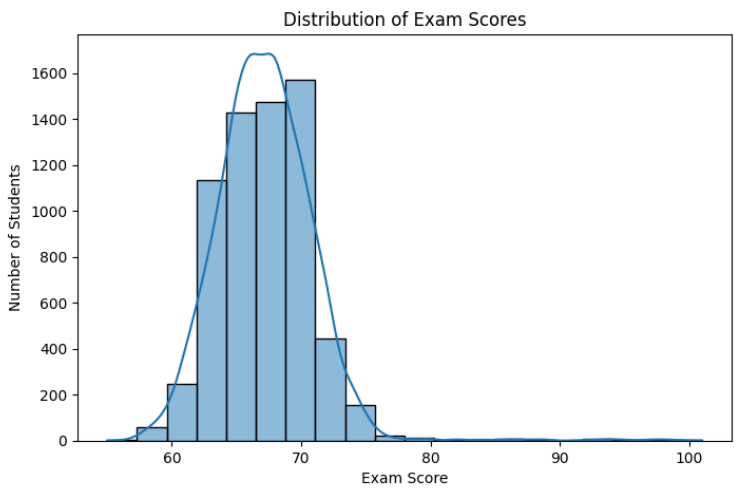

Figure 1: Distribution of Exam Scores

2.6.2 Feature Correlation

A correlation heatmap was generated to analyze relationships between features.

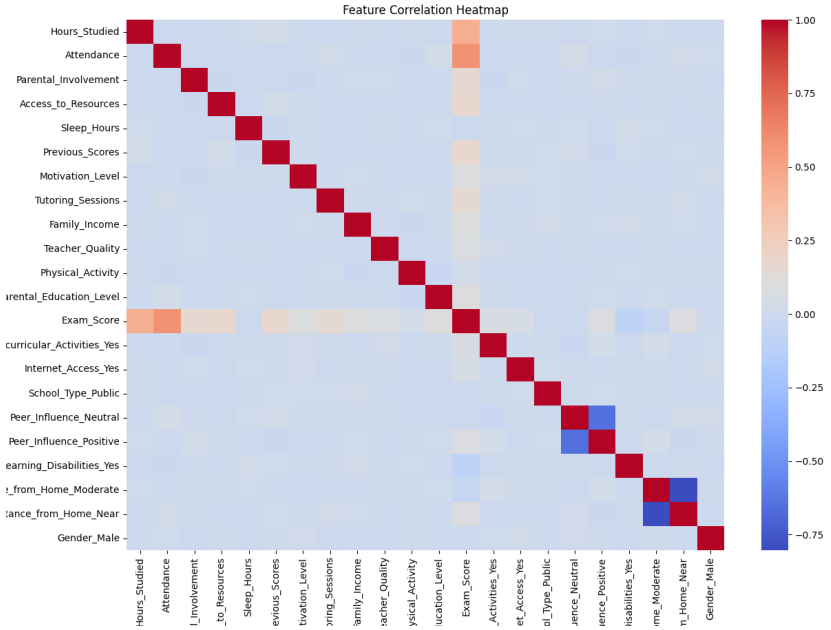

Figure 2: Correlation Heatmap of Dataset Features

2.6.3 Study Hours vs Exam Score

The following scatter plot shows the relationship between study hours and exam performance.

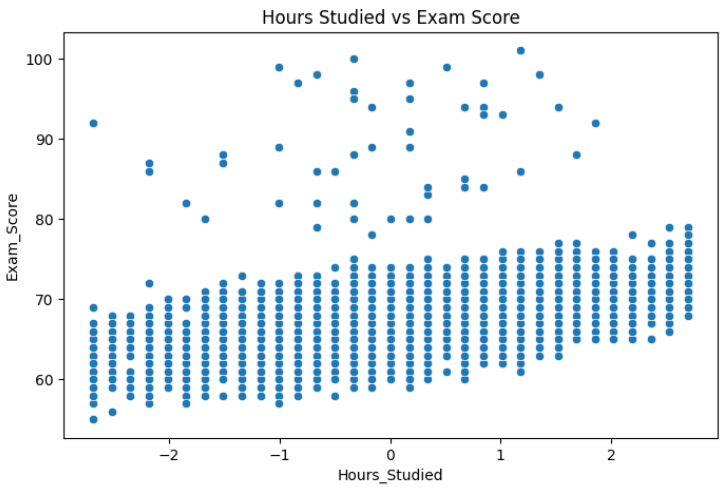

Figure 3: Relationship Between Study Hours and Exam Score

2.6.4 Sleep Hours vs Exam Score

Sleep patterns can influence cognitive performance. The following plot visualizes this relationship.

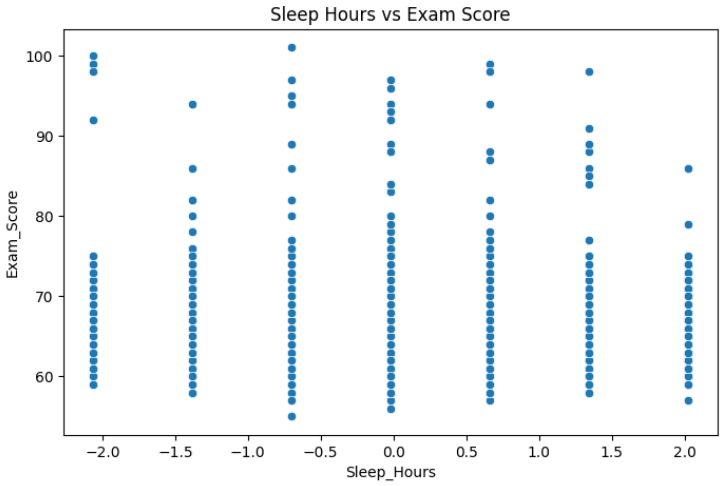

Figure 4: Relationship Between Sleep Hours and Exam Score

2.6.5 Motivation Level vs Exam Score

Motivation levels were compared with exam scores using a box plot.

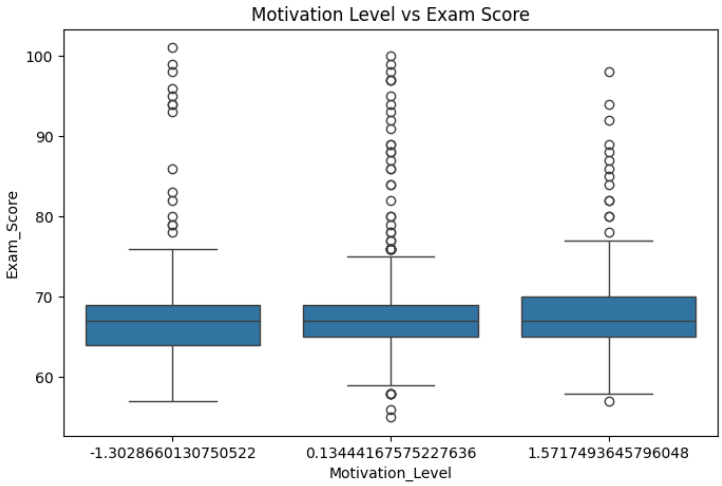

Figure 5: Motivation Level and Exam Score

3 Conclusion

In this deliverable, the foundational stages of the StudyPlan AI project were completed. Phase 1 defined the problem, target audience, and justification for using artificial intelligence. Phase 2 focused on acquiring a suitable dataset and performing extensive preprocessing to transform raw data into a structured format ready for machine learning analysis.

The cleaned and processed dataset now enables further development of predictive models that will estimate student exam performance and generate personalized study recommendations.